Aga Khan University Examination Board

Notes from E-Marking Centre on HSSC-II Chemistry Examination April/ May 2019

Introduction

This document has been produced for the teachers and candidates of Higher Secondary School Certificate (HSSC) Part II Chemistry. It contains comments on candidates' responses to the 2019 HSSC-II Examination, indicating the quality of the responses and highlighting their relative strengths and weaknesses.

E-Marking Notes

This includes overall comments on candidates' performance on every question and *some* specific examples of candidates' responses which support the mentioned comments. Please note that the descriptive comments represent an overall perception of the better and weaker responses as gathered from the e-marking session. However, the candidates' responses shared in this document represent some specific example(s) of the mentioned comments.

Teachers and candidates should be aware that examiners may ask questions that address the Student Learning Outcomes (SLOs) in a manner that requires candidates to respond by integrating knowledge, understanding and application skills they have developed during the course of study. Candidates are advised to read and comprehend each question carefully before writing the response to fulfil the demand of the question.

Candidates need to be aware that the marks allocated to the questions are related to the answer space provided on the examination paper as a guide to the length of the required response. A longer response will not in itself lead to higher marks. Candidates need to be familiar with the command words in the SLOs which contain terms commonly used in examination questions. However, candidates should also be aware that not all questions will start with or contain one of the command words. Words such as 'how', 'why' or 'what' may also be used.

General Comments

Candidates displayed better understanding of the concepts such as trends in physical and chemical properties of group I and VII elements, structures of alkyl halides, preparation of carboxylic acid by Grignard's reagent, condensation polymerisation, destructive distillation of coal, causes and impacts of smog, combustion analysis, preparation of derivatives of carboxylic acid and importance of dyes and pesticides. On the other hand, they showed weak understanding and need more focus on concepts; like, spectroscopic techniques, structures of aldehydes and ketones, purpose of iodoform test, electrophilic aromatic substitution reactions of phenol, mechanisms of elimination reaction of alkyl halides, nomenclature of coordination compounds, factors that affect enzyme activity and reactivity of л-bonds.

Detailed Comments:

Constructed Response Questions (CRQs)

Question 1ai:

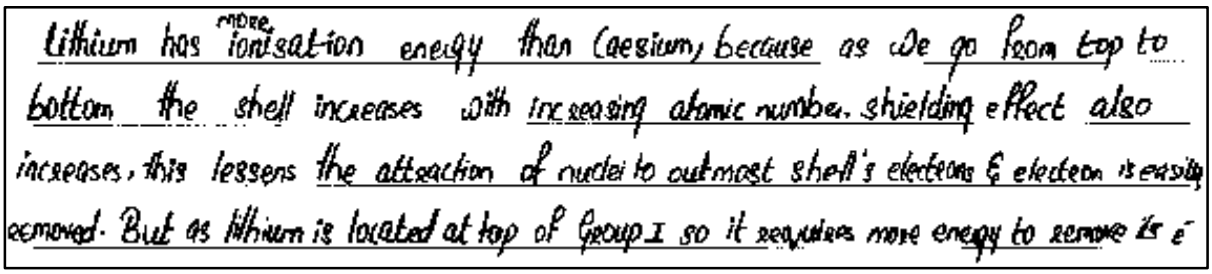
Which of the following metals has higher ionisation energy? Give reason to support your answer.

I. Caesium

II. Lithium

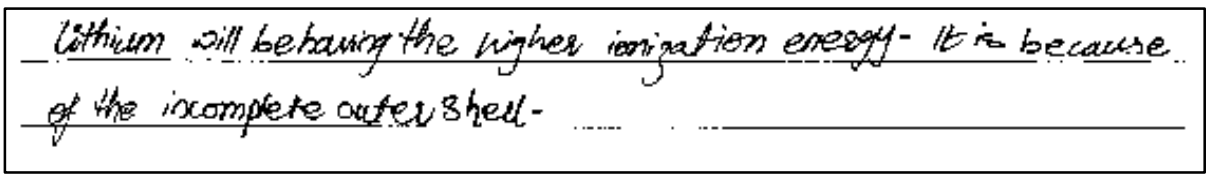
Better responses correctly identified lithium as the metal with higher ionisation energy. These responses justified their identification with reference to lithium placed above caesium in group I of the modern periodic table. Candidates mentioned that as we move down the group I of the modern periodic table, the nuclear charge increases. But as the size of the atom and the number of electrons causing the shielding effect also increases; therefore, ionization energy decreases from top to bottom.

Example:


Lithium has more ionisation energy than Caesium, because as we go from top to bottom the shell increases with increasing atomic number. shielding effect also increases, this lessens the attraction of nuclei to outmost shell's electrons & electron is easily removed. But as lithium is located at top of Group I so it requires more energy to remove its e

Weaker responses showed poor understanding of periodic trends of group I elements. A few of these candidates identified lithium correctly but failed to support their identification with logical reason. For example, the incorrect answers were lithium has higher ionisation energy because lithium belongs to group II A with oxidation of +2/ ionization energy is the energy to remove the most loosely bonded electron from the outer shell/ it has three electrons and only one electron in its valence shell so it will donate its electron and become ionised/ ionisation energy increases across the period.

Example:

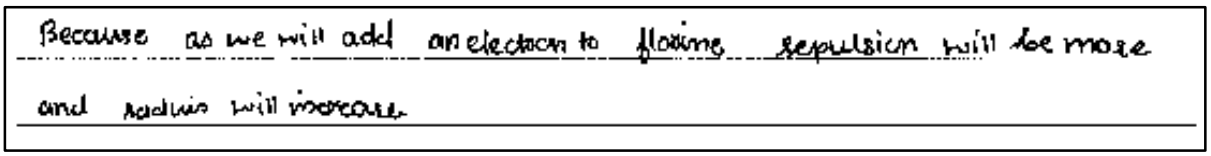

Lithium will be having the higher ionization energy- It is because of the incomplete outer shell-

Question 1aii:

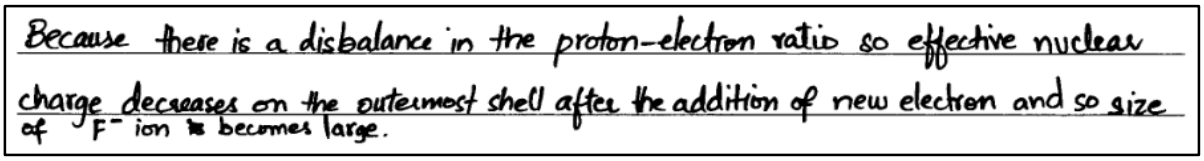
Why is the radius of fluorine atom smaller than that of the fluoride ion?

Better responses exhibited the clarity in justifying the difference between the radii of fluorine atom and its ion. The correct reasons included statements like addition of an electron in the shell of a neutral atom enhances repulsion between the electrons causing expansion of the shell. Thus, the radius of fluorine atom is less than that of the fluoride ion/ effective nuclear charge decreases and electron density increases in a fluoride ion/ fluorine atom has smaller radius due to more nuclear attraction between its nucleus and electrons.

Example 1:

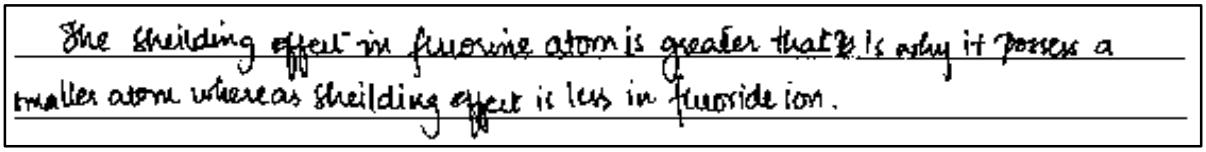


Example 2:



Weaker responses depicted lack of understanding of the difference in radii of fluorine atom and its ion. Candidates wrote incorrect statements that showed their misconceptions like due to shielding effect as fluorine atom has more electrons than fluorine ion so the attractiveness towards nucleus is more in fluorine than its ion/ fluorine atom has increased shielding effect/ radius of fluorine ion is greater due to higher electronegativity difference and greater shielding effect/ radius of fluorine atom is smaller because of the absence of unpaired electron and in fluoride ion, the orbital is partially filled.

Example:



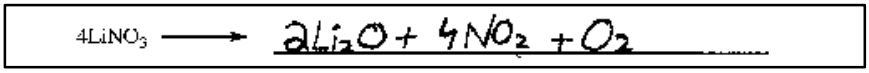
Question 1b:

Write the products formed when the given nitrate of lithium (Li) is decomposed.

$4LiNO_3 \longrightarrow$ _____

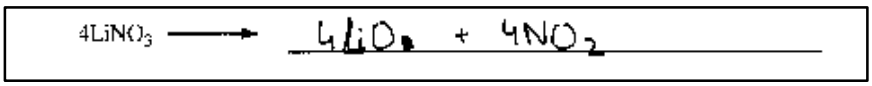
Better responses demonstrated clear understanding of the effect of heat on nitrates of group I elements. These responses provided correct balanced chemical equation by mentioning lithium oxide, nitrogen dioxide and oxygen as the products of the decomposition of lithium nitrate.

Example:

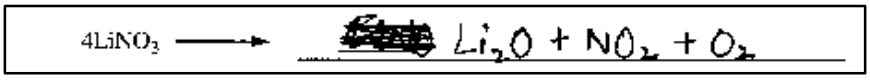


Weaker responses depicted errors either with reference to product formation or balancing. They wrote the correct products of the decomposition of lithium nitrate but didn't balance the equation or they formed incorrect products such as $4LiO + 4NO_2$/ $3NO_3$+ NO_3/ $4NO_2$ +LiO_4/ $2Li_2O + 3N_2O_3 + O$.

Example 1:



Example 2:



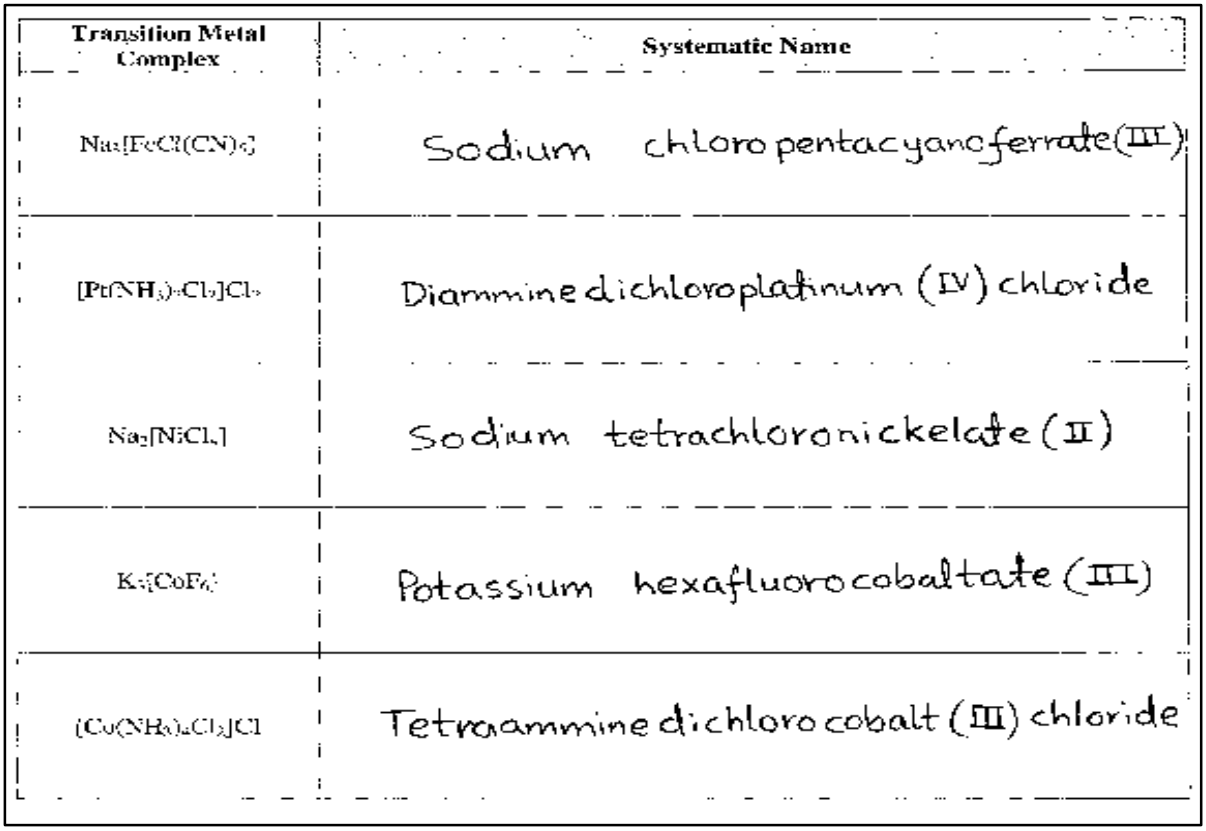
Question 2:

Write the systematic names of the given transition metal complexes.

Transition Metal Complex	Systematic Name
$Na_3[FeCl(CN)_5]$	
$[Pt(NH_3)_2Cl_2]Cl_2$	
$Na_2[NiCl_4]$	
$K_3[CoF_6]$	
$[Co(NH_3)_4Cl_2]Cl$	

Better responses provided the correct nomenclature of coordination compounds. These responses depicted candidates' strong grip over the rules to be followed while naming transition metal complexes. They ensured correct placement of oxidation states, suffixes, prefixes and names of ions and ligands.

Example:



Transition Metal Complex	Systematic Name
$Na_3[FeCl(CN)_5]$	Sodium chloro pentacyanoferrate(III)
$[Pt(NH_3)_2Cl_2]Cl_2$	Diammine dichloroplatinum (IV) chloride
$Na_2[NiCl_4]$	Sodium tetrachloronickelate (II)
$K_3[CoF_6]$	Potassium hexafluorocobaltate (III)
$[Co(NH_3)_4Cl_2]Cl$	Tetraammine dichloro cobalt (III) chloride

Weaker responses showed lack of grip on the rules of nomenclature. These responses showed errors in identifying the correct oxidation state, names and order of ligands and metal ions. A few candidates identified the names of $Na_2[NiCl_4]$ and $K_3[CoF_6]$ correctly. However, in rest of these responses, candidates wrote amino or ammonia in place of ammine/ chlorido instead of chloro/ iron rather than ferrate/ copper in place of cobalt. They even shuffled the order of ligands and mentioned names such as:

$Na_3[FeCl(CN)_5]$ = Sodium pentacyano chloro ferrate (III)

$[Pt(NH_3)_2Cl_2]Cl_2$ = Diammine dichlorine platinium (II) chlorine/ Diammonia dichloro platinium (IV) chloride

$Na_2[NiCl_4]$ = Sodium buta chloro nickel (II)

$K_3[CoF_6]$ = Potassium hexafluoro copperate (III)

$[Co(NH_3)_4Cl_2]Cl$ = Chloro pentaamino cobalt (III) chloride/ Buta ammine dichlorine cobult (I) chlorine

Example:

Transition Metal Complex	Systematic Name
$Na_3[FeCl(CN)_5]$	Sodium - chloro - penta-cyno - Ironate (iv)
$[Pt(NH_3)_2Cl_2]Cl_2$	Di-amine-di-chloro - Platenium - chloride
$Na_2[NiCl_4]$	Sodium - tetra chloro - Nickelate(ii)
$K_3[CoF_6]$	Potassium - hexaflouro - copperate (iii)
$[Cu(NH_3)_2Cl_2]Cl$	Tetra-amine-di-chloro - copper-chloride

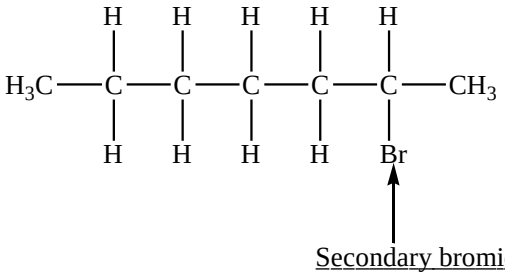
Question 3a:

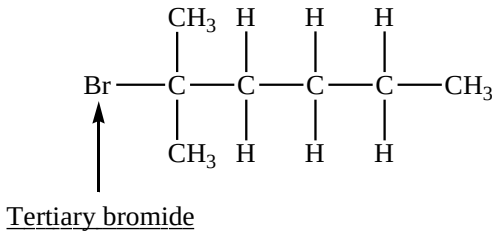
Draw the following structures with the molecular formula $C_7H_{15}Br$.

i. A secondary alkyl halide

ii. A tertiary alkyl halide

Better responses correctly drew the structures for the given alkyl halides using the molecular formula $C_7H_{15}Br$. These responses presented a variety of structures that depicted candidates' thorough understanding about the position of halogen atom in the alkyl chain of secondary and tertiary alkyl halides. For example,

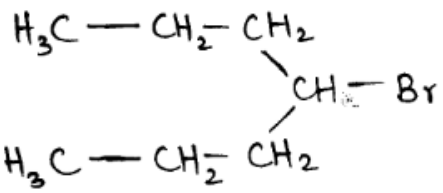




Example:

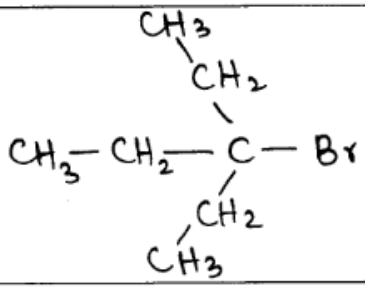
i. A secondary alkyl halide (1 Mark)

Space for drawing



ii. A tertiary alkyl halide (1 Mark)

Space for drawing

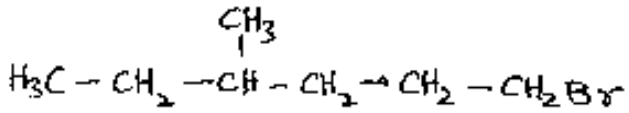


Weaker responses exhibited poor understanding about the structures of secondary and tertiary alkyl halides. A few of these responses figured out the correct structure of secondary alkyl halide but failed to draw the structure of tertiary alkyl halide. This seems as if the candidates based on the number of carbon, hydrogen and bromine atoms just randomly drew a chain in case of secondary alkyl halide and unknowingly placed the bromine atom at the right position. However, these responses didn't know the correct representation of tertiary carbon atom to which the bromide is attached in tertiary alkyl halide.

Example:

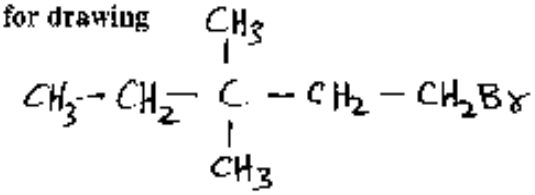
i. A secondary alkyl halide (1 Mark)

Space for drawing



ii. A tertiary alkyl halide (1 Mark)

Space for drawing



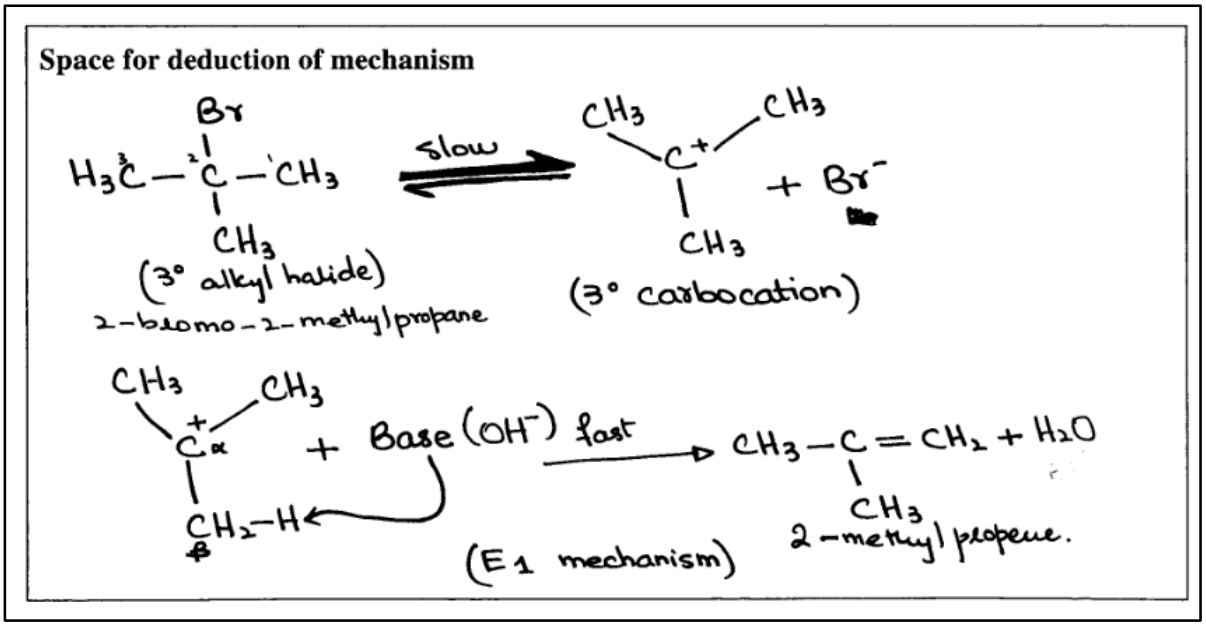
Question 3b:

2-bromo-2-methylpropane in the presence of alcoholic potassium hydroxide forms 2-methylpropene.

Deduce the mechanism of the given chemical reaction.

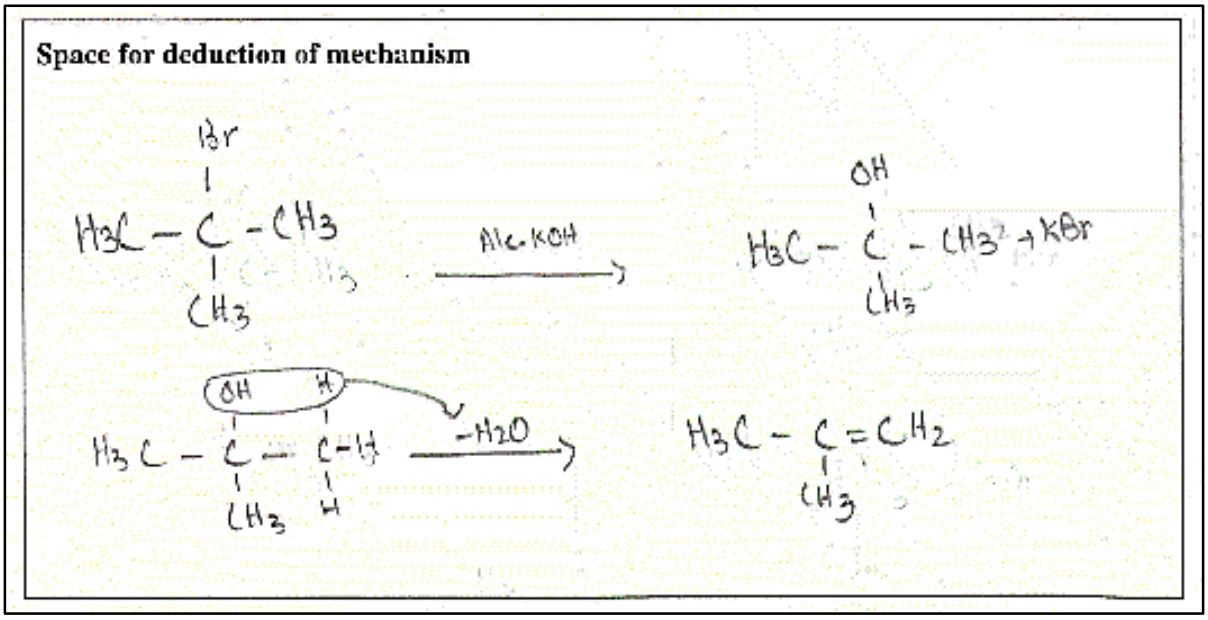
Better responses depicted in-depth understanding of E_1 mechanism. These responses showed the reaction in two steps. The first step presented the slow ionisation of the given substrate, i.e. 2-bromo-2-methylpropane giving a carbocation. In the second step, candidates showed the attack of nucleophile (OH^-) on hydrogen atom that is attached to the β-carbon atom in order to obtain the desired alkene as a product.

Example:



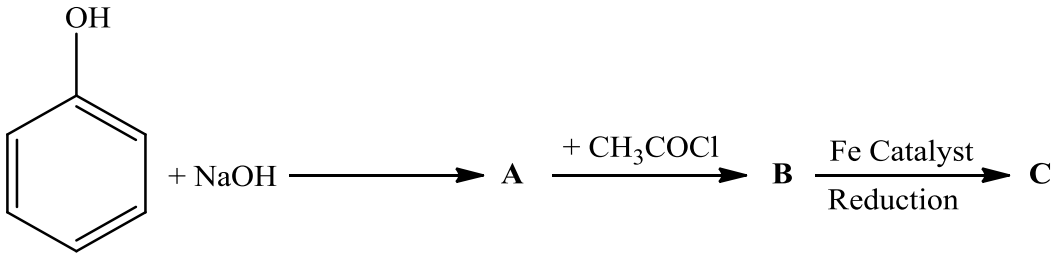
Weaker responses displayed two major mistakes in their answers. First, the candidates did not show the formation of carbocation. Secondly, instead of showing the attack of nucleophile on the hydrogen atom of β-carbon atom, candidates attached OH group to the tertiary (α) carbon atom. Moreover, candidates presented one-step equation of the complete reaction rather than showing the mechanism as per the demand of the question.

Example:



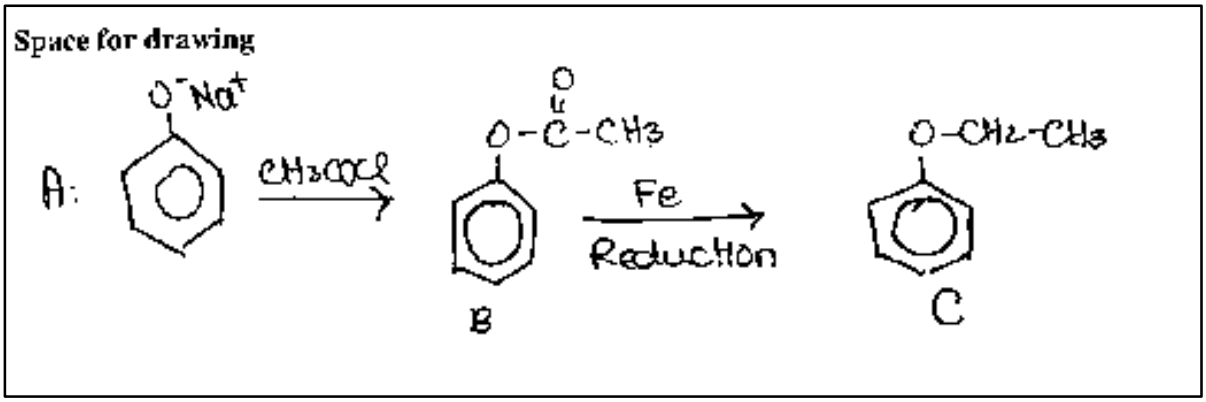
Question 4a:

Draw the structures of **A**, **B** and **C** to complete the given reaction of phenol.



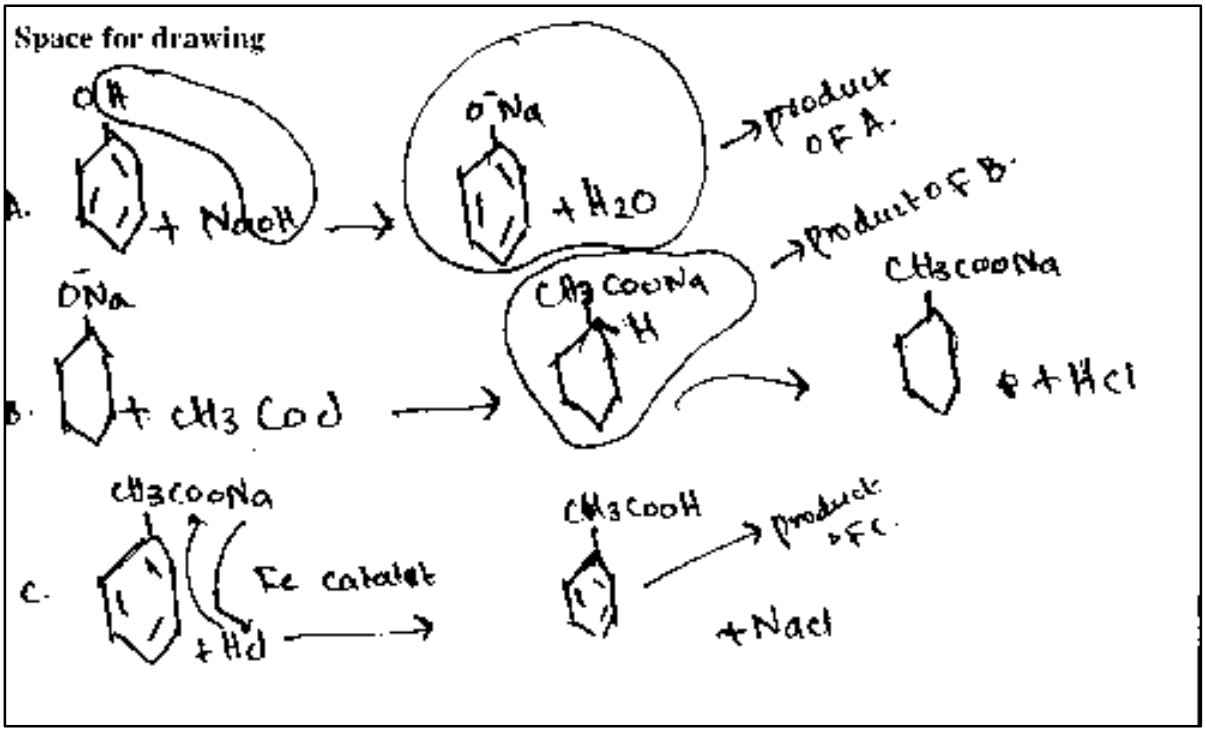
Better responses correctly identified the subsequent products, **A**, **B** and **C**, of the given reaction. Candidates drew the correct structures of sodium phenoxide ($NaOC_6H_5$), phenyl chloroformate (C_6H_5OCOCl) and ethoxy benzene/ ethyl phenyl ether ($C_6H_5OCH_2CH_3$).

Example:



Weaker responses depicted lack of knowledge about the electrophilic aromatic substitution reaction of phenol. These responses succeeded in drawing the structure of sodium phenoxide but struggled in predicting the subsequent structures of an ester and ether. The wrong structures showed direct attachment of CH_3COCl/ CH_3COONa to benzene ring in case of B and CH_3CHO/ CH_3COOH in case of C.

Example:

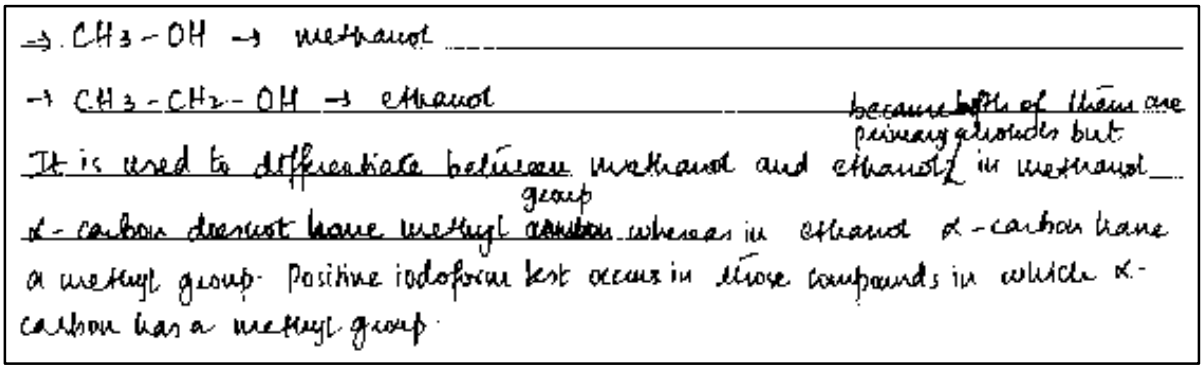


Question 4b:

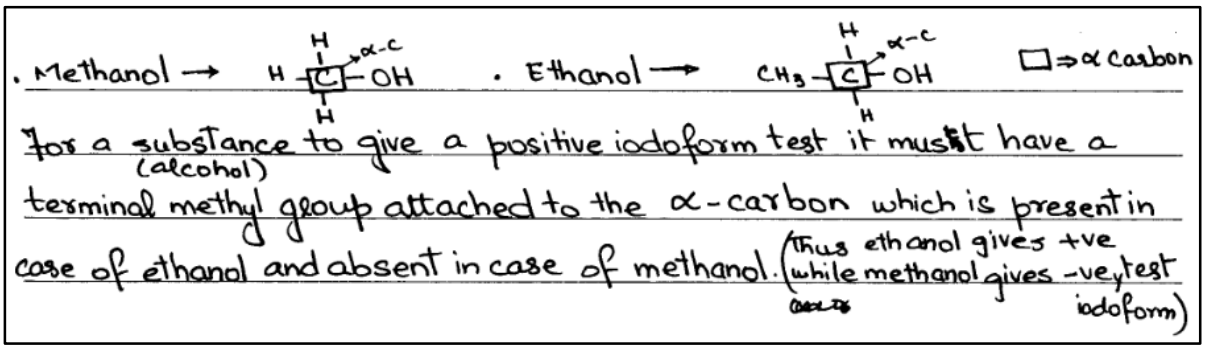
Why is iodoform test used to differentiate between methanol and ethanol? Give reason with reference to the structure of methanol and ethanol.

Better responses demonstrated a good grip over the purpose of using iodoform test to distinguish between specific alcohols. These responses correctly identified that iodoform test gives positive result for ethanol while it shows negative result for methanol because ethanol has a methyl group attached to the carbon with the -OH group.

Example 1:

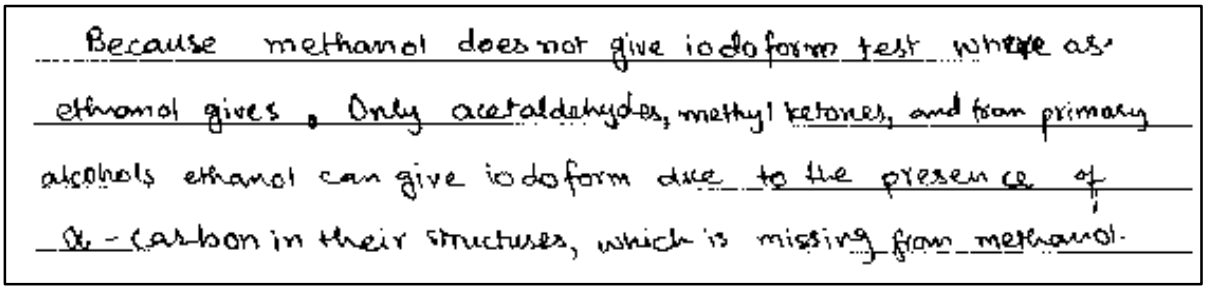


Example 2:



Weaker responses exhibited lack of understanding of the given test for alcohols. These responses could not identify the purpose of using iodoform test and produced responses with reference to ketones and aldehydes. Few responses correctly identified the test results, i.e. positive for ethanol and negative for methanol but the candidates could not give clear justification for this difference in result. They stated that α-carbon is present in ethanol and absent in methanol. The carbon attached to the functional group is termed as α-carbon which is actually present in both structures (CH_3CH_2OH and CH_3OH) – the difference is of methyl group which the candidates could not highlight. Their irrelevant responses included statements like methanol is less bulkier than ethanol which has more boiling point than methanol, also oily in nature, less volatile than methanol/ methanol gives positive test and ethanol gives negative test because those compounds can give positive iodoform test in which CH_3 is directly attached to the functional group/ iodoform test is used to differentiate between methanol and ethanol because methanol gives yellow crystal in iodoform test while ethanol do not give any crystal.

Example:

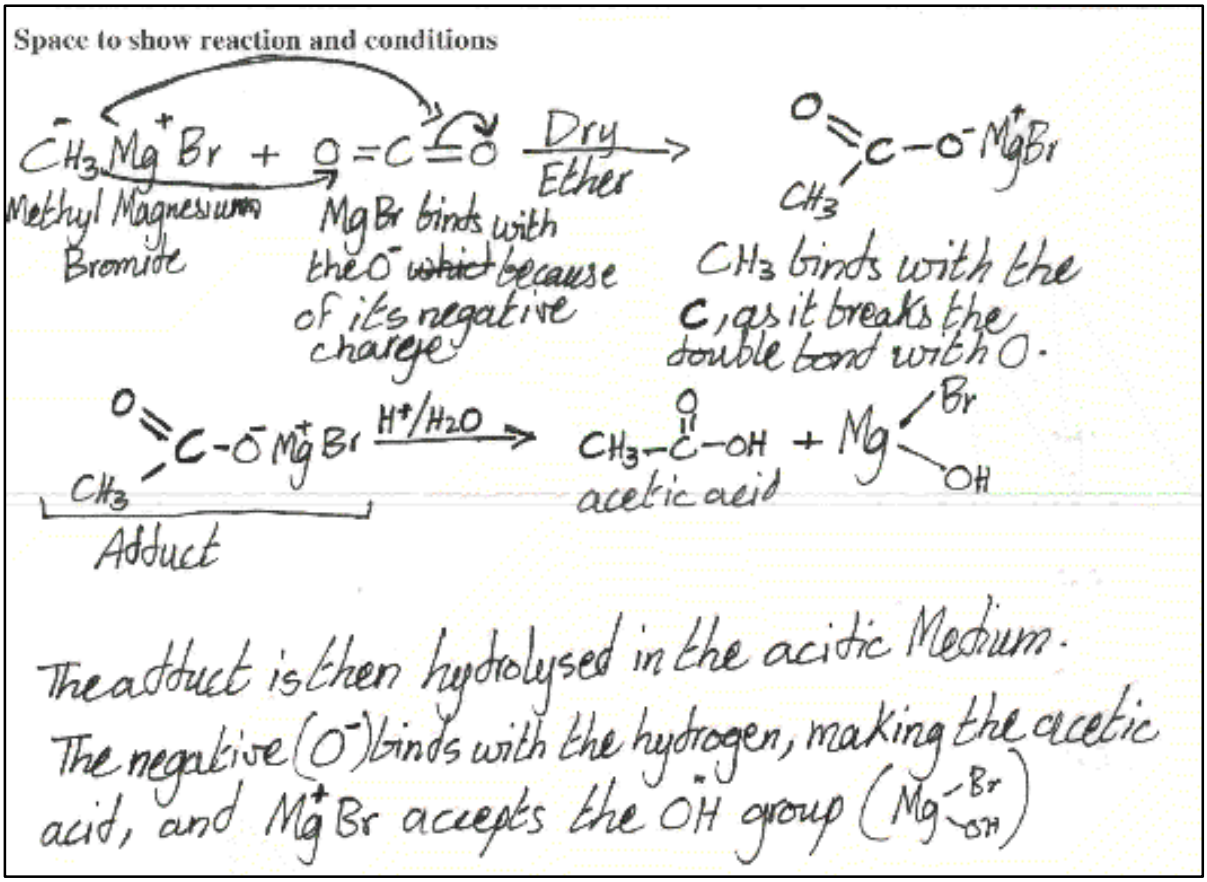


Question 5:

Show the preparation of acetic acid by reacting methylmagnesium bromide (Grignard's reagent) with carbon dioxide. Mention the conditions necessary for this reaction.

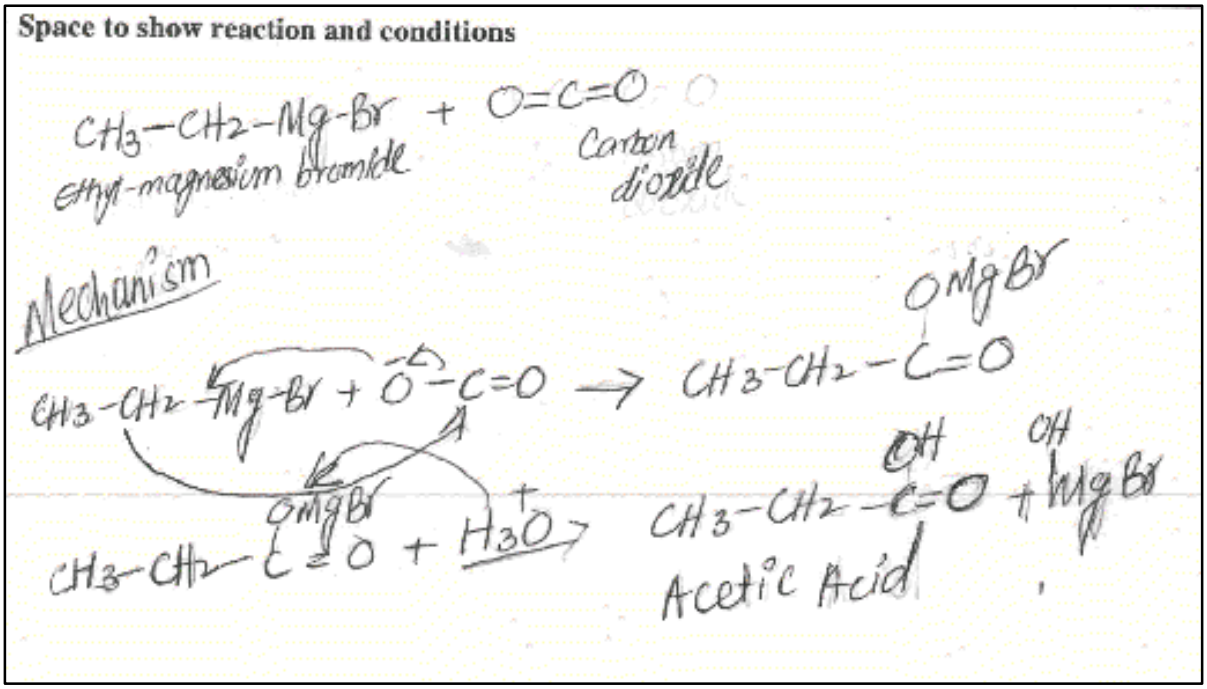
Better responses correctly displayed the preparation of acetic acid from the given reactants. These responses showed the formation of intermediate as well as mentioned dry ether to proceed the reaction followed by hydrolysis in acidic medium.

Example:



Weaker responses showed poor understanding about the preparation of carboxylic acid by Grignard's reagent. These responses depicted errors such as candidates did not know the formula of methylmagnesium bromide instead, they mentioned the structure of ethylmagnesium bromide; conditions were not specified; ether was used for the preparation of methylmagnesium bromide but in reaction its involvement was not shown; the intermediate [RCOOMgBr] was directly converted into acetic acid without reacting it with mineral acid or showing acidic hydrolysis on the arrow.

Example:



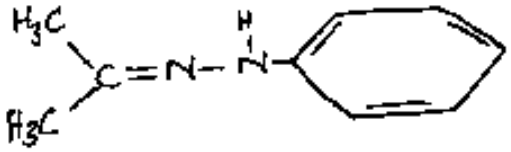
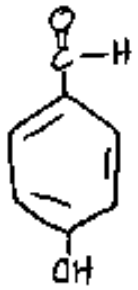
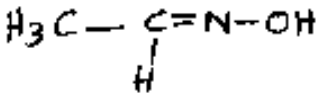
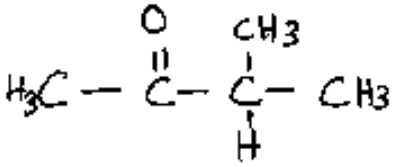
Question 6:

Draw the structural formulae of the following compounds.

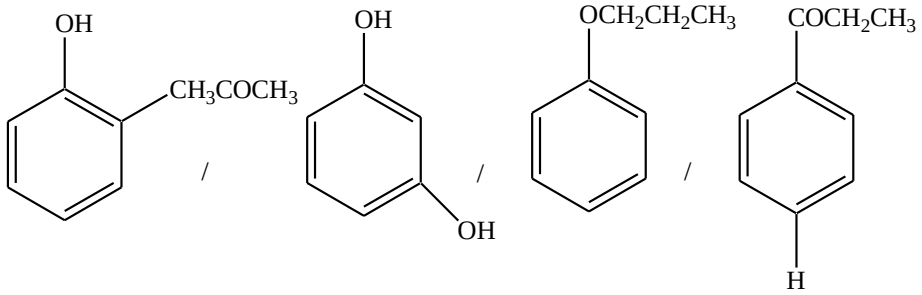
Compound	Structural Formula
Propanone phenylhydrazone	
4-hydroxybenzaldehyde	
Ethanaloxime	
3-Methyl-2-butanone	

Better responses demonstrated clarity in drawing structures of aldehydes, ketones and products of acid catalysed addition reactions of aldehydes and ketones. Candidates had clear understanding of the position and placement of functional groups including types of bonds.

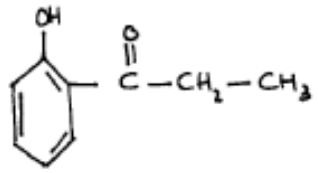
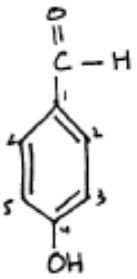
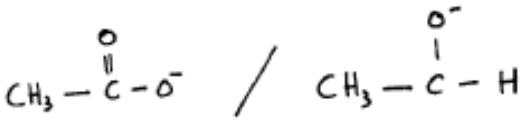
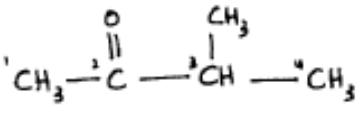
Example:

Propanone phenylhydrazone	
4-hydroxybenzaldehyde	
Ethanaloxime	
3-Methyl-2-butanone	

Weaker responses were unable to draw the structures of the given organic compounds. A few of these responses succeeded in identifying the correct structure of 4-hydroxybenzaldehyde or 3-methyl-2-butanone but these responses could not figure out all four structures. The incorrect structures candidates drew for 1 and 3 were as follows:

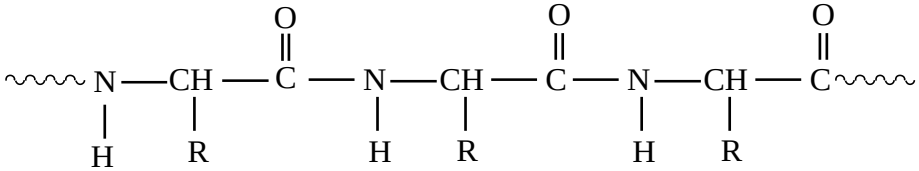
Propanone phenylhydrazone	
Ethanaloxime	CH_3CHO/ $CH_2(OH)CH_2(NH_2)$/ $CH_3CH_2CH_2NH_2OH$/ CH_2O-CHO

Example:

Propanone phenylhydrazone	
4-hydroxybenzaldehyde	
Ethanaloxime	
3-Methyl-2-butanone	

Question 7:

A section of protein is shown below.



i. Circle the peptide linkage in the given section of protein.

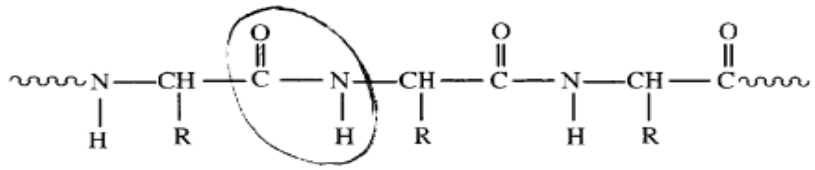
ii. Draw the structural formula for the monomer used to make the given protein.

iii. Describe the type of polymerisation process involved in the formation of the given protein.

Better responses depicted thorough understanding regarding the structure of protein and type of reaction involved in its making. In part 'i', these responses correctly circled CONH to specify the peptide linkage, drew $H_2N - CH(R) - COOH$ in part 'ii' to represent the structural formula of the monomer of the given polymer and finally, described condensation polymerisation in part 'iii' as the type of process that enables the making of protein through the elimination of a water molecule.

Example:

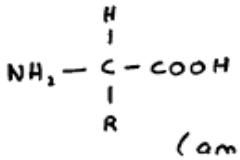
A section of protein is shown below.



i. Circle the peptide linkage in the given section of protein. (1 Mark)

ii. Draw the structural formula for the monomer used to make the given protein. (1 Mark)

Space for drawing



(aminoacid).

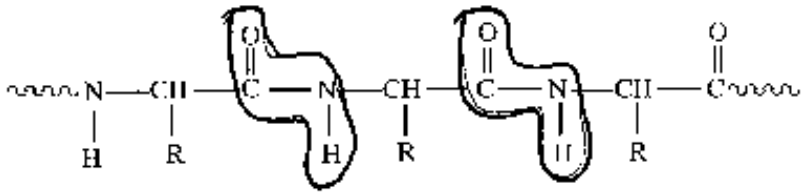
iii. Describe the type of polymerisation process involved in the formation of the given protein. (2 Marks)

The type of polymerisation is condensation polymerization in which small molecules (monomers) join to form larger molecules with the release of small compounds such as water (H_2O) or methanol (CH_3-OH) e.g water is removed in protein synthesis and nylon 6,6 production.

Weaker responses succeeded in identifying the peptide linkage in the given section of protein in part 'i' but failed to attempt part 'ii' and 'iii'. In part 'ii', these responses drew incorrect structures of amino acid and identified odd/ incorrect type of polymerisation process such as giant/ synthetic/ addition/ natural/ homopolymerisation. Instead of specifying the R attached to CH group candidates mostly showed CH_2 group in amino acid (monomer) or wrote the formula as **COOH**CH_2NH_2 or **NH**$_2CH_2COOH$ which shows H bonded to carbon in both bold cases. In part 'iii', the irrelevant statements candidates produced were addition polymerisation is involved to form the protein because of small polymers are combine to form the protein/ giant polymerisation is used in which many polypeptide linkages are made through peptide bond between amino acid to make the complex protein molecule/ polymerisation is process in which small monomers are combined together to make a monomer and then lot monomer joined to form protein/ polymerisation process is homopolymerisation as same type of monomers (in this case amino acids) combine to form massive, long chain polymer and macromolecule (in this case).

Example:

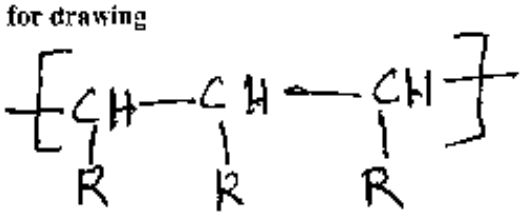
A section of protein is shown below.



i. Circle the peptide linkage in the given section of protein. (1 Mark)

ii. Draw the structural formula for the monomer used to make the given protein. (1 Mark)

Space for drawing



iii. Describe the type of polymerisation process involved in the formation of the given protein. (2 Marks)

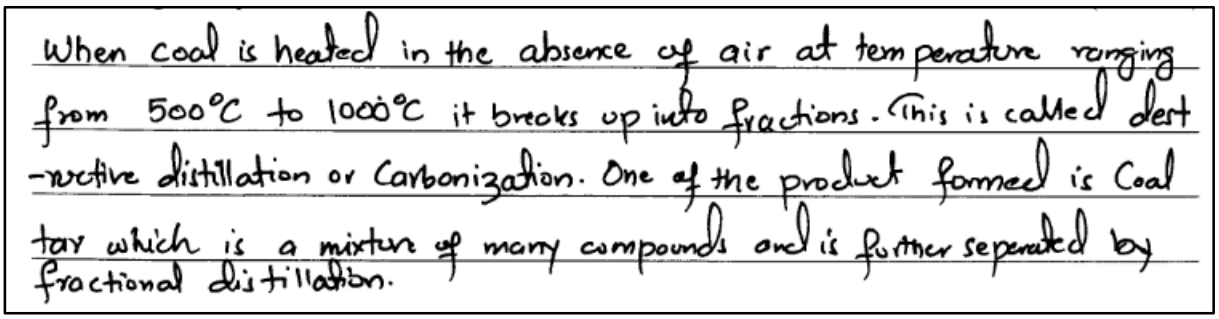
Addition polymerisation: It is a type of polymerisation in which two same molecules combine together forming long polymers and the simple and same monomers cause addition polymerisationsuch as PVC (polyvinyl chloride).

Question 8a:

Describe the process of destructive distillation of coal and name any ONE product obtained through this process.

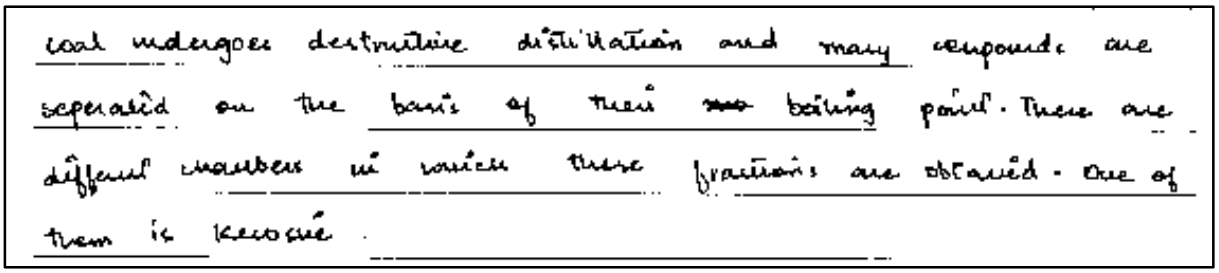
Better responses described the process of destructive distillation in detail. These responses highlighted the basic requirements, i.e. absence of air and temperature ranging from 500-1000°C at which coal is heated to produce many by-products. Names of various by-products were observed in such responses. For example, coke, coal tar, ammonical liquor, carbon gas and coal gas.

Example:


When coal is heated in the absence of air at temperature ranging from 500°C to 1000°C it breaks up into fractions. This is called dest-ructive distillation or Carbonization. One of the product formed is Coal tar which is a mixture of many compounds and is further seperated by fractional distillation.

Weaker responses exhibited lack of knowledge regarding destructive distillation of coal. Candidates mixed the given concept with fractional distillation and mentioned separation of coal at different boiling points/ destructive distillation of coal is a process in which coal is burned at high pressure in order to generate by-products/ coal is heated at high temperature and pressure to get products. However, few of these responses were able to name the correct product obtained as a result of this process.

Example:


coal undergoes destructive distillation and many compounds are seperated on the basis of their ~~~~ boiling point. There are different numbers of carbon these fractions are obtained. One of them is kerosene.

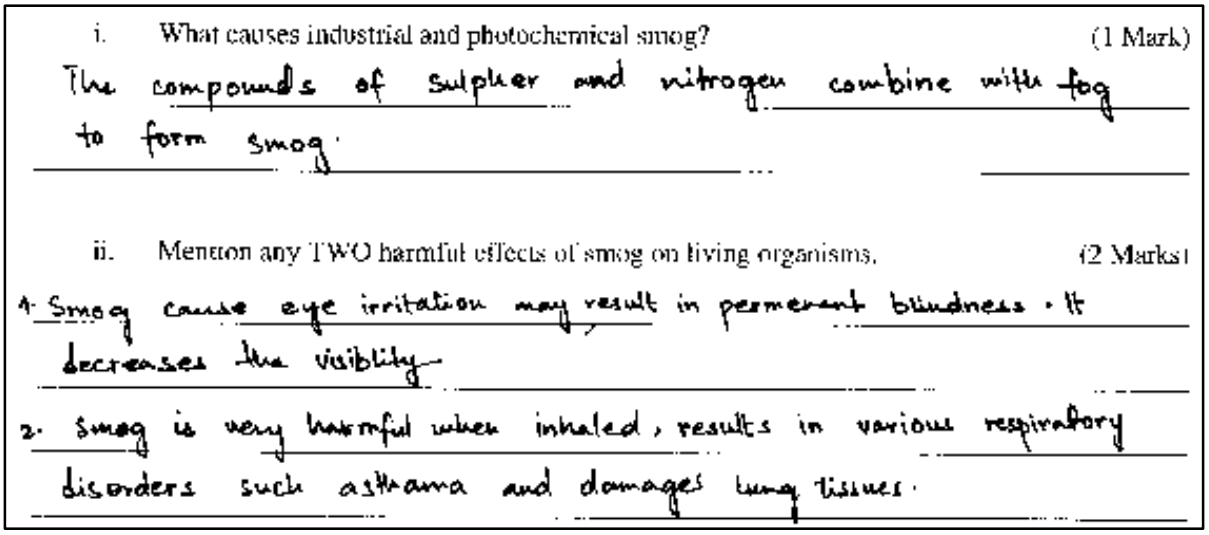
Question 8b:

i. What causes industrial and photochemical smog?

ii. Mention any TWO harmful effects of smog on living organisms.

Better responses showed a clear understanding about the cause and effect of smog. Candidates mentioned sulphur oxides/ sulphur bearing fossil fuels/ coal fires/ reaction of nitrogen oxides with sunlight/ reaction of nitrogen oxides and volatile organic compounds (hydrocarbons) with sunlight/ vehicular and industrial emissions/ volcanic eruptions as the cause of industrial and photochemical smog. Furthermore, their correct effects included smog causes or aggravate asthma/ emphysema/ chronic bronchitis and other respiratory problems/ eye irritation/ reduced resistance to colds and lung infections.

Example:



i. What causes industrial and photochemical smog? (1 Mark)

The compounds of sulpher and nitrogen combine with fog to form smog.

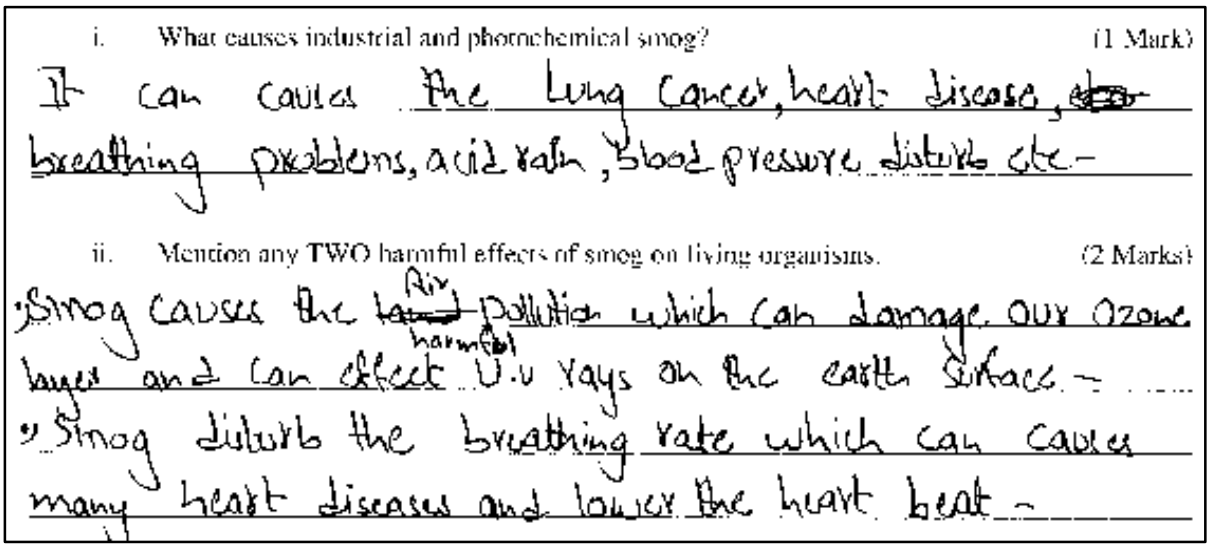
ii. Mention any TWO harmful effects of smog on living organisms. (2 Marks)

1. Smog cause eye irritation may result in permenent blindness. It decreases the visiblity.

2. Smog is very harmful when inhaled, results in various respiratory disorders such asthama and domages lung tissues.

Weaker responses demonstrated poor understanding about the causes and effects of smog. Candidates wrote effects of smog in part 'i' and 'ii'. These responses included all sorts of illogical effects such as heart diseases/ blood pressure/ acid rain/ damages the ecosystem/ pollutes water/ smog is skin harming, atmosphere harming in which organisms take breathe while smog pollutes/ smog is harmful solvent giving rise to diseases when is in exposure to living organisms.

Example:



i. What causes industrial and photochemical smog? (1 Mark)

It can cause the lung cancer, heart disease, breathing problems, acid rain, blood pressure disturb etc.-

ii. Mention any TWO harmful effects of smog on living organisms. (2 Marks)

1. Smog causes the ~~layer~~ Air pollution which can damage our Ozone layer and can effect U.V rays harmful on the earth surface.-

2. Smog disturb the breathing rate which can causes many heart diseases and lower the heart beat.-

Question 9a:

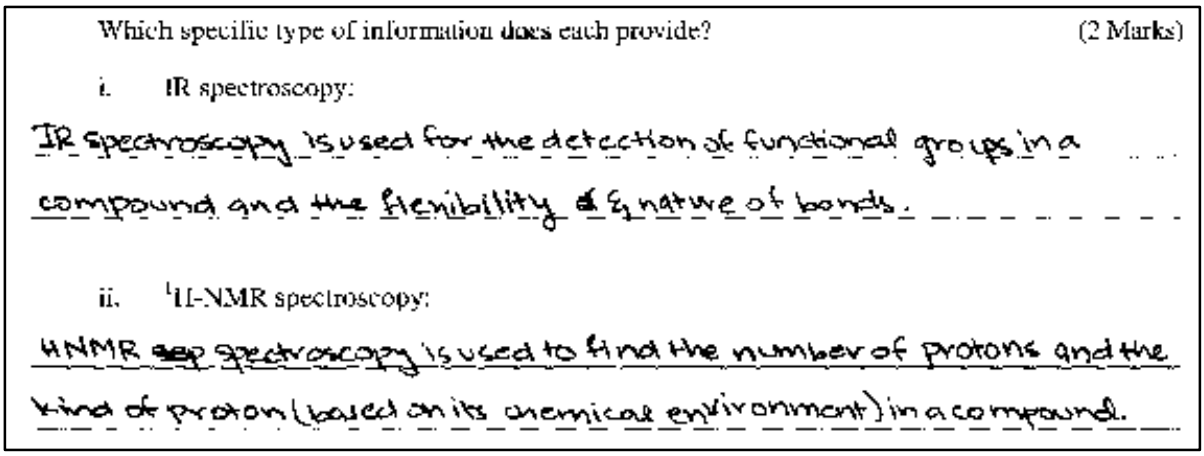
Infrared (IR) and proton nuclear magnetic radiation (^{1}H-NMR) spectroscopy are both used to identify structures of molecules.

Which specific type of information does each provide?

i. IR spectroscopy:
ii. ^{1}H-NMR spectroscopy:

Better responses mentioned the specific information that is studied in infrared and proton nuclear magnetic radiation spectroscopy. These responses showed students interest and clarity in concepts related to analytical chemistry. Candidates mentioned all related information, for example, IR spectroscopy provides information about the type of functional groups present in an unknown compound/ vibrations inclusive of bending ad stretching. Similarly, ^{1}H-NMR spectroscopy is useful for identifying the number and type of hydrogen atoms as well as position of carbon atoms in a molecule. They also wrote about nuclear spin, spin-spin coupling and chemical shift.

Example:


Which specific type of information does each provide? (2 Marks)

i. IR spectroscopy:

IR spectroscopy is used for the detection of functional groups in a compound and the flexibility & nature of bonds.

ii. ^{1}H-NMR spectroscopy:

HNMR aap spectroscopy is used to find the number of protons and the kind of proton (based on its chemical environment) in a compound.

Weaker responses demonstrated lack of knowledge about the information one can retrieve from IR spectroscopy and ^{1}H-NMR spectroscopy. A few of these responses showed correct answer to part 'i' but struggled to give the right answer of part 'ii'. Candidates found it easier to recall that IR spectroscopy technique is quite helpful for identifying the functional groups/ type of bond in an unknown compound but mentioned incomplete statement about ^{1}H-NMR spectroscopy that it provides information about proton. What exactly of proton was not specified such as it gives information about the 'number/ type/ environment of hydrogen atoms in a molecule'. Their vague and irrelevant responses included statements like ^{1}H-NMR spectroscopy determines radicals in the molecule/ it gives information about weight/ it gives information about the double bond present in a molecule.

Example:

Which specific type of information does each provide? (2 Marks)

i. IR spectroscopy:

The vibration of bond give information that functional group is present.

ii. ^{1}H-NMR spectroscopy:

It give information about weight.

Question 9b:

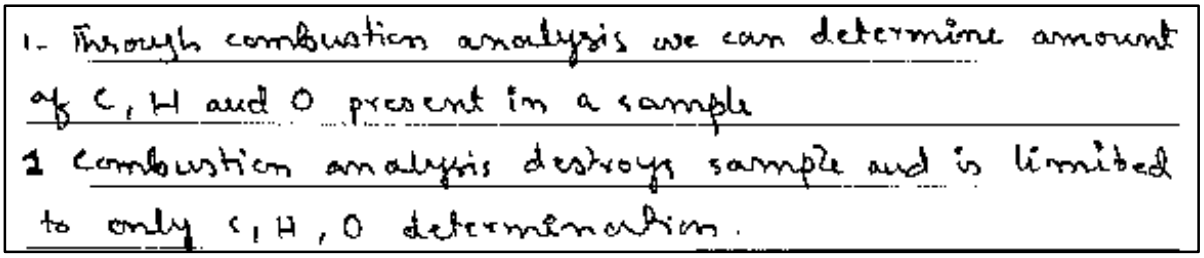
State ONE use and ONE drawback of combustion analysis.

Better responses managed to state one use and one drawback of combustion analysis. The uses that candidates mentioned clearly specified the benefits of combustion analysis technique such as it determines the elemental composition (more precisely empirical formula) of a pure organic compound/ the relative amounts of carbon, hydrogen, and oxygen in the starting compound can be determined from the masses of the products of the combustion reaction/ it is used to find out the composition of organic compound, i.e. the amount of hydrogen, carbon and oxygen present in it. These responses also depicted correct drawback of combustion analysis such as combustion analysis is only limited to those organic compounds which contain carbon, hydrogen and oxygen/ it does not find composition of elements other than carbon, hydrogen and oxygen/ sample gets destroyed and cannot be reused.

Example 1:

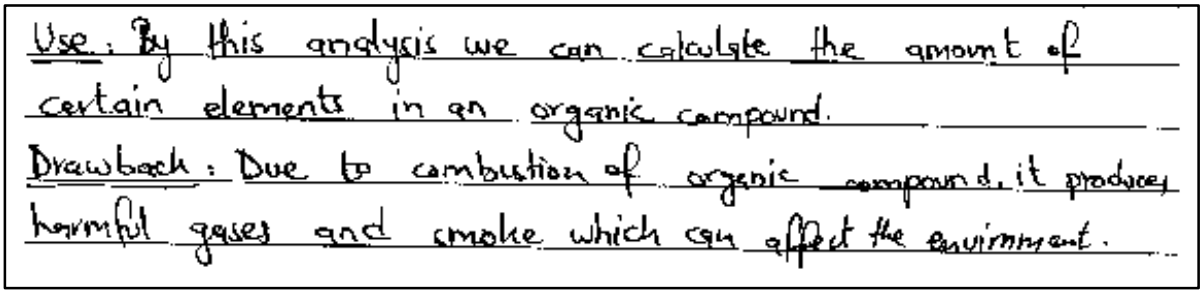
• Combustion analysis is used to identify empirical formula and hence molecular formula of an organic compound.
• Its one draw back is that the sample is finished and can not be reused.

Example 2:



Weaker responses exhibited lack of understanding about combustion analysis. These responses produced incorrect use and drawback of combustion analysis. The incorrect responses included statement such as combustion analysis is used to identify the amount of molecule in a sample/ it is used to split components of a compound or material by combustion/ it is used to get the desired product in a short period of time. With reference to drawback, candidates' misconceptions were observed. They mixed combustion analysis technique with combustion reaction that takes place in the atmosphere. In combustion analysis, the combustion products are secured for quantitative analysis in order to determine the amount of carbon, hydrogen and oxygen from which the empirical formula is further calculated. In contrast, candidates stated that combustion analysis highly contributes to air pollution/ most of the components or derivatives evaporate in the form of vapours.

Example:



Extended Response Questions (ERQs)

These questions offered a choice between part **a** and **b**.

Most of the candidates attempted question 10b and performed better compared to question 10a. Though question 10b was based on organic reactions, but candidates interpreted the reactions well using balanced chemical equations. Similarly, question 11b was attempted the most by candidates and they performed better compared to question 11a. It seemed candidates were not much familiar with the effects of various factors on enzyme activity in question 11a.

Question 10a:

i. Explain why alkenes are more reactive towards electrophilic reagents than alkynes.

ii. Write a chemical equation showing preparation of alkenes from

I. dehydrohalogenation of alkyl halide.

II. dehydration of alcohol.

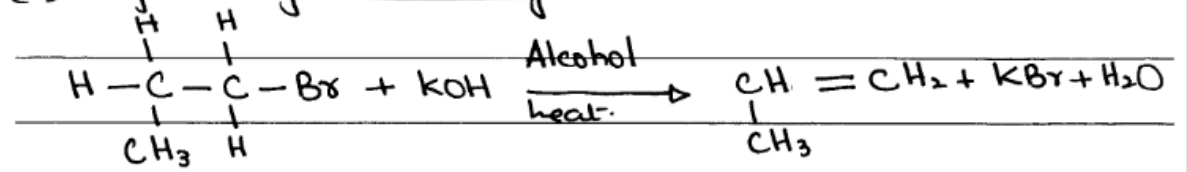
(**Note**: In the equations, mention all the necessary conditions for reactions.)

Better responses, in part 'i', well explained the difference between alkene and alkyne in terms of reactivity towards electrophilic reagents. In these responses, candidates showed the clarity of concept, i.e. in case of alkene, π-electrons are less firmly held between the nuclei which makes the π-bond weak in comparison to the σ-bond. Whereas, in alkynes, the carbon atoms are held together by a triple bond, a σ-bond and two π-bonds. The electron density between the carbon atoms in alkynes is high which draws atoms close to each other. During the reaction with an electrophilic reagent, the loosely held π-electrons in alkenes are more exposed to attack by the electrophilic reagents in contrast to alkynes and therefore the π-bond breaks comparatively easily rendering alkenes as more reactive than alkynes. Furthermore, candidates in part 'ii' showed the correct preparation of alkenes from dehydrohalogenation of alkyl halide and dehydration of alcohol. These responses depicted appropriate conditions to carry-out the given reactions such as heating of alkyl halide in the presence of alcoholic potassium hydroxide and use of catalyst like heated alumina/ concentrated sulphuric acid were mostly mentioned for dehydrating alcohols.

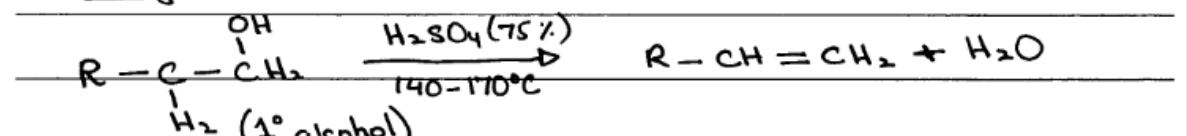
Example:

In alkene functional group, each carbon of the C=C is sp^2 hybridized. It is linked with the other carbon via a sigma(σ) and a pi(π) bond. In the pi-bond the maximum probability of finding an electron is above and below the line joining the two nuclei. Thus, pi-bond electrons are easily available for attack because pi-bond is a weak bond and it can easily break and electrons are available for reaction as they are not held very tightly between the two nuclei. Thus alkenes are very reactive towards electrophilic reagents. While, in alkynes the carbon atom are sp hybridized (C≡C) and each carbon forms a sigma bond (sp-sp overlap) and 2 pi-bonds by the overlapping of two unhybridized p-orbitals of each C-atom. As a result of an extra pi-bond, the bond length between C≡C atoms in alkynes (below that of alkene) decreases due to which the pi-bond electrons are held very tightly between the two nuclei and are not easily available for an attack by electrophiles. Thus, due to this the reactivity of alkynes is less than that of alkenes. (They have higher bond strength than alkenes).

I. Dehydrohalogenation of alkyl halide:



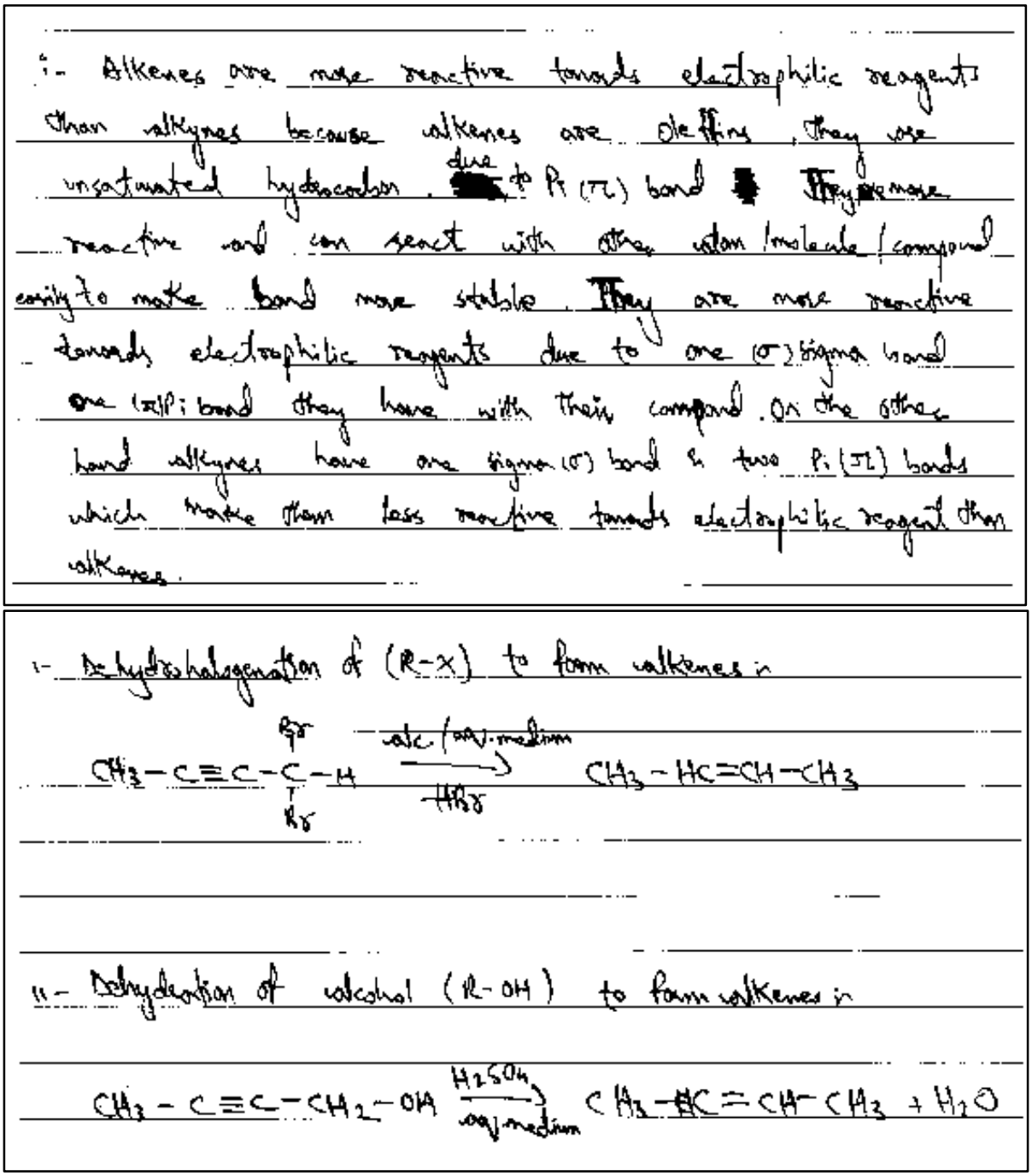
II Dehydration of alcohol:



Dehydrating agents: H_2SO_4, P_4O_{10}, Al_2O_3.

Weaker responses exhibited poor understanding about the difference in reactivity of a π-bond in alkenes and alkynes. Candidates produced incorrect responses in part 'i' as well as 'ii'. These responses depicted the difference in the number of π-bonds in alkenes and alkynes but could not elaborate upon the way the π-electrons are held or exposed or take part in reactions. In part 'ii', instead of choosing alkyl halide and alcohol as the substrate for preparing alkene; candidates displayed incorrect compounds such as alkynes/ alkanes. These responses also depicted choice of wrong conditions inclusive of medium/ base/ catalyst for the given reactions.

Example:



Question 10b:

Using balanced chemical equations,

i. show the preparation of propanoyl chloride by reacting propanoic acid with thionyl chloride in the presence of pyridine.

ii. show the reaction of propanoyl chloride with

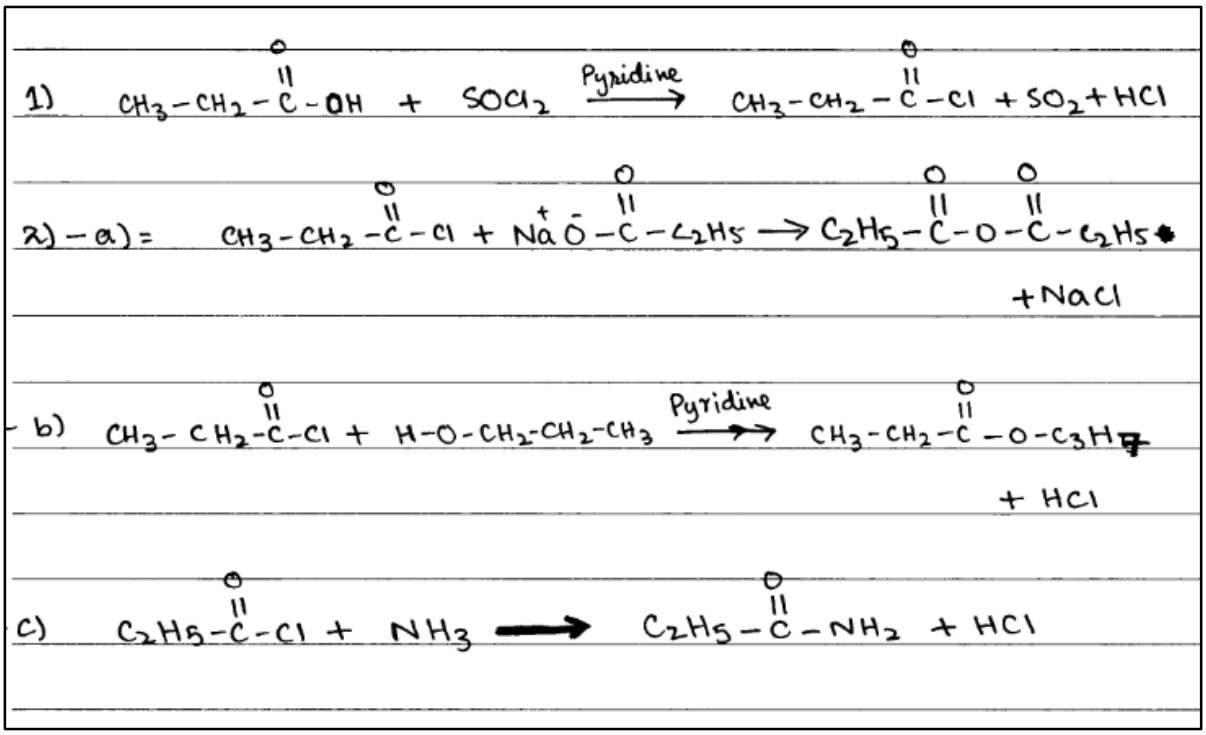
I. sodium propanoate to distill over propanoic anhydride.

II. propanol in the presence of pyridine to prepare propyl propionate.

III. ammonia to form propanamide.

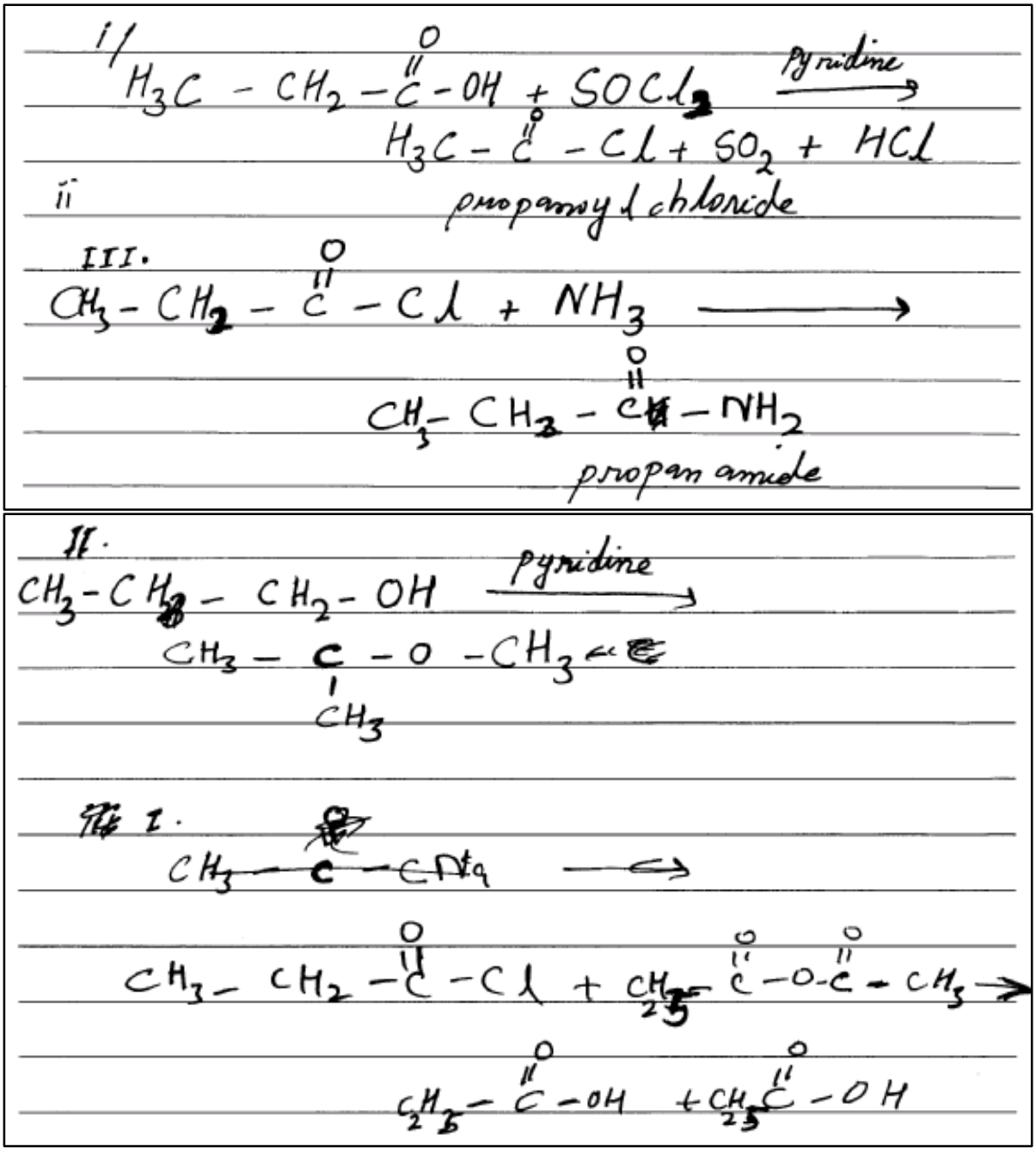
Better responses depicted strong grip over the concept of carboxylic acids and their derivatives. Candidates, using the given reactants, correctly showed the preparation of the desired acyl halide, acid anhydride, ester and amide. The correct representation of the structure of reactants and products was quite evident in majority of the responses. In few responses, candidates all showed the mechanism of the reactions which exhibited their interest towards organic reactions.

Example:



Weaker responses showed confusion in representing the correct reactants and products in the given question through chemical equation. These responses either represented the first equation for preparation of propanoyl chloride correctly or succeeded in preparing the given amide. These responses mostly made error in depicting the correct structure of propanoyl chloride and thus, continued further given reactions with the wrong structure. Instead of propanoic acid, few candidates showed the reaction with butanoic acid which depicted candidates rush to answer without thorough reading and understanding of the question.

Example:



Question 11a:

i. Why are enzymes called biological catalysts?

ii. Explain the effects of the following factors on enzyme activity:

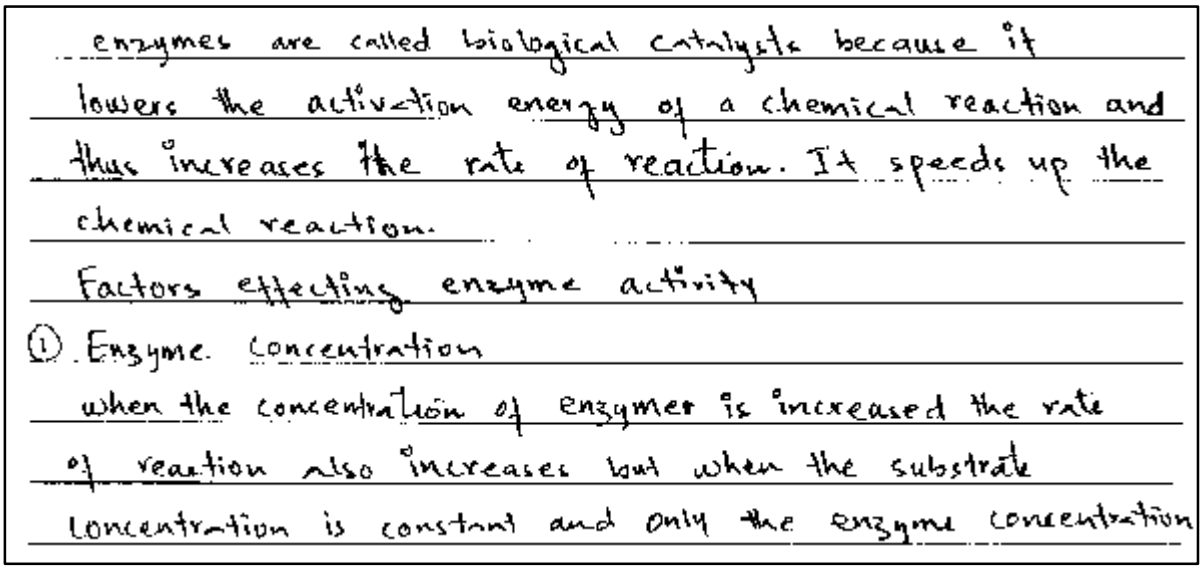
I. Enzyme concentration

II. Temperature

III. pH

Better responses correctly explained the reason for naming enzymes as a biological catalyst in part 'i'. These responses mentioned that similar to catalysts, enzymes accelerate the rate of chemical reactions and they are neither consumed nor changed in the reaction/ enzymes are proteins functioning as catalyst that speed up reaction by lowering the activation energy. In part 'ii', candidates explained the effects of each of the given factors on enzyme activity. For example, in case of enzyme concentration, these responses highlighted the effect with reference to increase and decrease in the concentration of enzyme and availability of substrate. In terms of temperature, candidates explained about the effect of increased, decreased and optimum temperature on enzyme activity. Similarly, these responses further elaborated upon the effect of pH on enzyme activity referring to change in its value and related examples that depicted working in optimum environment. Candidates also specified about extreme changes in temperature and pH can cause enzymes to denature and permanently lose their function.

Example:



enzymes are called biological catalysts because it lowers the activation energy of a chemical reaction and thus increases the rate of reaction. It speeds up the chemical reaction.

Factors effecting enzyme activity

① Enzyme concentration

when the concentration of enzymes is increased the rate of reaction also increases but when the substrate concentration is constant and only the enzyme concentration

is increased the reaction enzyme will proceed in constant manner because there will be no substrate to get attached with enzyme.

2- Temperature

Every enzyme has an optimum temperature at which they become active. when temperature increase the enzyme activity also increases but when the temperature is increased above the optimum temperature, the enzymes get denatured and the reaction stops there.

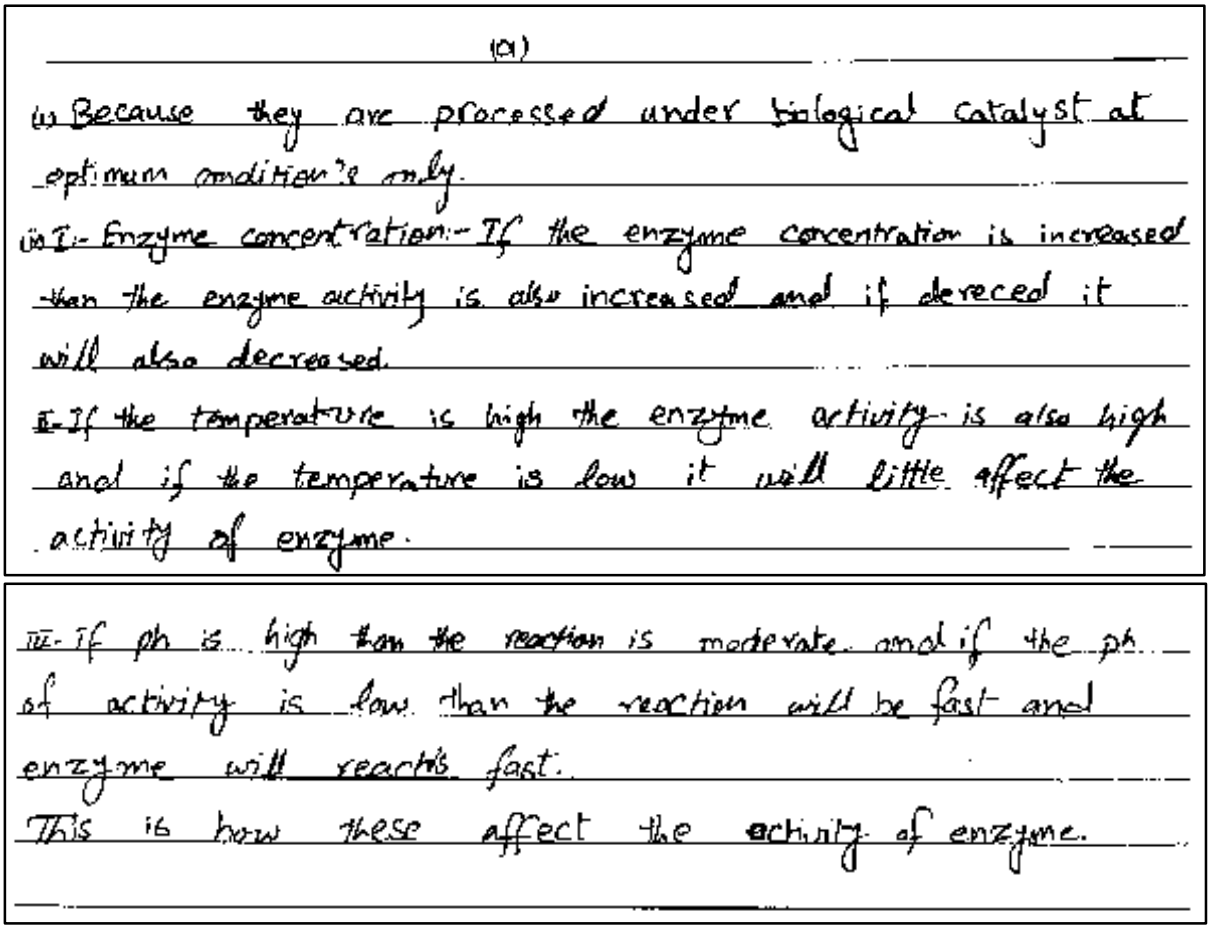


3- PH

PH also effects the enzyme activity. every enzymes works at different PH. some are activated in acidic medium and some in basic medium. If the enzymes which needs acidic medium is placed in basic medium it will not be activated like Pepsinogen - it needs acidic medium for it's activation.

Weaker responses presented the correct reason for enzymes to be biological catalysts but failed to explain the effects of the given factors on enzyme activity. These responses mostly had similar statements such as with increasing enzyme concentration, temperature and pH the enzymes activity will increase and by lowering the activity would decrease. Candidates did not have the understanding of optimum conditions and denaturing of enzyme/ retardation of enzyme activity by increasing or decreasing a certain factor. For example, candidates wrote that if temperature is high then the enzyme activity will be high and when it is low then it will have little effect on the activity of enzyme.

Example:



(a)

(i) Because they are processed under biological catalyst at optimum condition's only.

(ii) I- Enzyme concentration:- If the enzyme concentration is increased then the enzyme activity is also increased and if decreced it will also decreased.

II- If the temperature is high the enzyme activity is also high and if the temperature is low it will little affect the activity of enzyme.

III- If ph is high than the reaction is moderate and if the ph of activity is low than the reaction will be fast and enzyme will reacts fast.

This is how these affect the activity of enzyme.

Question 11b:

i. Write the importance of each of the given pesticides:

   I. Molluscicides

   II. Rodenticides

   III. Herbicides

   IV. Fungicides

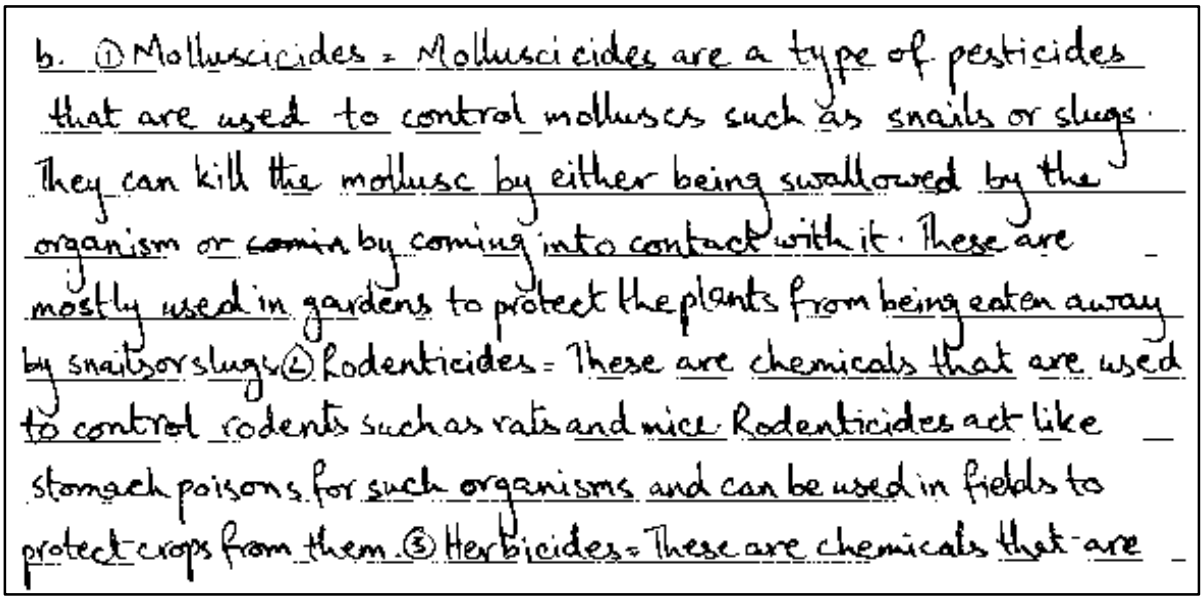
ii. What are dyes?

iii. What is the colour of methyl orange in acidic medium and basic medium?

Better responses constructed the explanation in sequence. In part 'i', these responses correctly described the importance of each of the given pesticides with reference to their role in controlling different kinds of pests. For example, molluscicides are used to control snails and slugs; rodenticides control rats, mice, bats and other rodents; herbicides control unwanted plants and fungicides control the fungi which cause molds, rots, and plant diseases.

Furthermore, these responses shared the understanding of candidates regarding dyes such as dyes are coloured compounds which are used in solution form and are capable of being fixed to a fabric. The dyes must be fast or chemically stable so that the colour will not wash with soap and water, or fade on exposure to sunlight (ultraviolet light). These responses also identified the correct change of colour of methyl orange, i.e. red in acidic medium and yellow in basic medium.

Example:


b. ① Molluscicides = Molluscicides are a type of pesticides that are used to control molluscs such as snails or slugs. They can kill the mollusc by either being swallowed by the organism or coming into contact with it. These are mostly used in gardens to protect the plants from being eaten away by snails or slugs ② Rodenticides = These are chemicals that are used to control rodents such as rats and mice. Rodenticides act like stomach poisons for such organisms and can be used in fields to protect crops from them. ③ Herbicides = These are chemicals that are

used to control the growth of unwanted plants. Herbicides are of two types; non-selective herbicides that are toxic to all plants and selective herbicides that kill or suppress the growth of the target plants. This is used to prevent the uptake of minerals and nutrients by such plants.
@ Fungicides. These are chemicals that are used to control the growth of fungus. They kill or inhibit the growth of fungi by coming into contact with them as fungi do not swallow in the normal sense. These are also used to prevent the utilization of minerals by fungi so that they can be saved for other crops or plants and to prevent the plants from getting fungal diseases.
ii DYES: Dyes are coloured chemical substances mostly in the form of a solution that have the ability of being fixed to a fabric. Dyes get their colouring property from chromophores and their fixing property from auxochromes present in their chemical structure. Methyl orange is a type of azo dye that turns red in acidic medium and yellow in basic or alkaline medium. ....

Weaker responses exhibited lack of understanding about pesticides and dyes. Fungicides and herbicides were common to a few for which candidates mentioned the significance correctly. However, they failed to identify the importance of molluscicides and rodenticides. Furthermore, definition of dye was presented in general terms and the colour change of methyl orange was identified in either of the two medium.

Example:

b(i). impatance of each given pesticide:

iii. Herbicides : it is used to kill germs and bacterias from crops and herbs.

iv. Fungicides : these pesticides are used to kill or destroy Fungus or fungi material ~~more~~ . these are also used in medicines.

ii. Rodenticides : these are the type of pesticides which are used to remove infectious bacterias and harmfull germs.

i. Malluscicides : these are used for medicinal purposes as well as in crops whose soil is not so much fertile & rarely it is also used for roofing systems.

II. Dyes : these are the type of material which (solvents) colorize other colored or uncolored materials, inothes words dyes are the compounds used for coloring purposes. (dyes are mixture of many different elements or compounds).

III. color of methyl orange in acidic medium is:

= dark orange. (nearest to red but still orange)

. in basis medium it is :

= yellow